Experiment 17

Experiment to understand Linear Regression for a given data set

Aim:

Understand the Linear regression for the dataset given.

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df

df.info()

df.dropna(inplace=True)

df.info()

df.describe()

features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=0)

from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
model.fit(x_train,y_train)

model.score(x_train,y_train)

model.score(x_test,y_test)

model.coef_

model.intercept_

import pickle
pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
▾ LinearRegression
LinearRegression()
```

```
0.9603182547438908
```

```
    Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

```
0.9184170849214232
```

```
array([[9281.30847068]])
```

```
array([27166.73682891])
```

Result:

A Simple Linear Regression model was successfully trained to predict employee salary from years of experience. The model achieved an R2 score of approximately 0.96, indicating a strong positive linear relationship. The trained model was serialized and saved for future inference.